Estimation, Specification, and Computational Assessment of the Order Acceptance Function

Table 1: Estimated coefficients and significance levels of the bounded nonlinear regression model

Variable	**Coefficient**	***p*-value**
Constant	0.950	3.977×10^{-88}
Waiting_Time	−0.015	4.461×10^{-7}
Discount	0.138	3.719×10^{-2}
Detour_Ratio	−0.408	8.998×10^{-50}
Pickup_Time	−0.015	3.855×10^{-8}
Normal_Trip_Minutes	−0.004	4.874×10^{-7}
Pickup_Time×Discount	0.011	4.519×10^{-2}
Normal_Trip_Minutes×Discount	0.004	4.000×10^{-2}
Detour_Ratio×Discount×Waiting_Time	0.050	1.312×10^{-3}
R^2	0.772 (Train), 0.720 (Validation)	
MSE	0.023 (Train), 0.026 (Validation)	

Although the coefficients are estimated by OLS, the adopted acceptance function contains multiple interaction terms involving discount, pickup time, detour ratio, waiting-for-matching time, and normal trip duration. It is therefore nonlinear in the original explanatory variables. In addition, the fitted function is bounded to the probability interval $[0, 1]$. We hence refer to the specification used in the integrated optimization model as the bounded nonlinear regression specification.

0.1 Alternative Behavioral Specifications

We compare the bounded nonlinear regression specification with exact Fractional Logit using the same explanatory variables, interaction terms, estimation sample, and train–validation split.

Table 2: Statistical fitting comparison of the order acceptance functions

Model	Validation MSE	Validation R^2
Unbounded OLS regression	0.026068	0.720300
Bounded nonlinear regression	0.025971	0.721338
Exact Fractional Logit	0.024581	0.736256

Relative to the bounded nonlinear regression specification, exact Fractional Logit reduces the validation MSE from 0.025971 to 0.024581, a reduction of 5.35%, and increases R^2 from 0.721338 to 0.736256, an increase of approximately 0.015. The improvement in predictive fit is therefore modest.

Binary Logit is additionally evaluated as a complementary sensitivity specification. Fractional Logit retains all 23,938 response records, whereas Binary Logit excludes 4,293 uncertain responses and therefore retains 19,645 records for estimation. Binary coding additionally converts 12,630 "very likely" responses from their ordered-response representation into binary outcomes.

On the common probability-valued validation target, Fractional Logit obtains an MSE of 0.024581 and an R^2 of 0.736256, compared with 0.030829 and 0.669220 for Binary Logit. On the retained binary validation records, Fractional Logit obtains a Brier score of 0.193645, log loss of 0.573982, AUC of 0.699941, and accuracy of 0.717512, compared with 0.197932, 0.590596, 0.695298, and 0.706452 for Binary Logit, respectively.

These results support using Fractional Logit as the principal nonlinear benchmark because it retains the probability-valued coding of all five response levels described in the probability-mapping appendix

of the main paper and provides better validation performance under the common data processing and train–validation partition. Binary Logit remains a reasonable complementary sensitivity specification for the constructed binary response.

0.2 Piecewise-Linear Approximation and Computational Implications

To embed Fractional Logit into the optimization model, we construct piecewise-linear approximations under seven requested maximum probability-error levels, ranging from 0.200 to 0.001. The requested maximum error specifies the approximation tolerance used to construct the piecewise-linear logistic function, whereas the verified maximum error is the largest probability approximation error subsequently measured over the relevant function domain. These approximation errors are different from the out-of-sample prediction MSE.

Table 3: Calibration of the piecewise-linear Fractional Logit approximations

Requested maximum probability error	Verified maximum probability error	Segments per logistic function	Validation MSE	MSE relative to bounded nonlinear
0.200	0.180497	2	0.031051	+19.56%
0.150	0.138311	3	0.027723	+6.74%
0.140	0.138311	3	0.027723	+6.74%
0.136	0.043662	4	0.023742	−8.58%
0.050	0.043662	4	0.023742	−8.58%
0.010	0.009129	9	0.024466	−5.80%
0.001	0.000672	32	0.024564	−5.42%

The calibration reveals a discrete change in approximation complexity between the three-segment and four-segment representations. The requested maximum error of 0.136 is therefore used as a threshold-near approximation that achieves at least comparable predictive accuracy to the bounded nonlinear regression specification, while 0.010 is retained as a finer approximation closer to the exact logistic curve.

The prediction MSE is not expected to decrease monotonically with the maximum piecewise-linear approximation error. The former evaluates predictions against the validation responses, whereas the latter evaluates the numerical approximation of the fitted logistic function. Accordingly, the approximation error is used to control functional fidelity rather than as a monotonic control on prediction MSE.

0.3 Computational Effects in O-CAM

We next embed all seven Fractional-Logit approximations into the O-CAM formulation. We use ten `A_4_2` instances and ten `A_6_3` instances and report the results under a common 7,200-second time limit. All specifications are evaluated in the same computational environment. A run terminates when Gurobi returns an `OPTIMAL` status or when the time limit is reached.

To avoid comparing an observed solution time with a time-limit-censored result, computational efficiency is compared only on instances for which both the bounded nonlinear regression and the corresponding Fractional-Logit formulations obtain `OPTIMAL` status. For each approximation setting, the runtime ratio is calculated by dividing the mean runtime of the Fractional-Logit formulation by the mean runtime of the bounded nonlinear regression formulation over the same jointly optimal instances. Instances for which the bounded nonlinear regression formulation obtains `OPTIMAL` status but the Fractional-Logit formulation reaches the time limit are reported separately.

All `A_4_2` formulations obtain Gurobi `OPTIMAL` status. The bounded nonlinear regression formulation requires 4.95 seconds on average, whereas the seven Fractional-Logit approximations require between

Table 4: O-CAM results under alternative acceptance-function specifications

Acceptance model	A_4_2 (10 instances)			A_6_3 (10 instances)				
	Mean time (seconds)	Runtime ratio	Optimal / time-limit	BNR optimal / time-limit	Logit optimal / time-limit	BNR mean time (seconds)	Logit mean time (seconds)	Runtime ratio
Bounded nonlinear regression	4.95	1.00	10 / 0	10 / 0	–	970.71	–	–
Logit, error 0.200	18.69	3.78	10 / 0	10 / 0	4 / 6	304.44	3,164.99	10.40
Logit, error 0.150	21.92	4.43	10 / 0	10 / 0	8 / 2	558.82	4,005.72	7.17
Logit, error 0.140	22.01	4.45	10 / 0	10 / 0	8 / 2	558.82	4,060.83	7.27
Logit, error 0.136	21.97	4.44	10 / 0	10 / 0	8 / 2	558.82	4,165.81	7.45
Logit, error 0.050	18.32	3.70	10 / 0	10 / 0	7 / 3	819.65	4,144.14	5.06
Logit, error 0.010	6.62	1.34	10 / 0	10 / 0	9 / 1	853.75	2,187.82	2.56
Logit, error 0.001	20.99	4.24	10 / 0	10 / 0	8 / 2	685.22	3,771.67	5.50

Note: BNR denotes the bounded nonlinear regression formulation. For A_4_2, all ten instances are jointly optimal, and the runtime ratio is the Logit mean runtime divided by the BNR mean runtime. For each Logit specification in A_6_3, the BNR and Logit mean runtimes and their ratio are calculated only over the instances for which both formulations obtain Gurobi OPTIMAL status. The BNR row reports its mean runtime over all ten A_6_3 instances.

6.62 and 22.01 seconds. Every Fractional-Logit approximation is slower than the bounded nonlinear regression formulation on each of the ten paired instances.

To examine whether this result is driven by only a few unusually slow instances, we additionally apply a one-sided paired sign test. The test considers only the direction of the runtime difference within each paired instance and tests whether the Fractional-Logit formulation is more likely to be slower than the bounded nonlinear regression formulation. Because the Fractional-Logit formulation is slower in all ten paired instances, the corresponding one-sided probability under the null hypothesis of equal likelihood is$(1/2)^{10} = 0.00098$. Thus, the observed computational disadvantage is consistent across the paired instances rather than being driven by a small number of extreme runtimes.

At the threshold-near error of 0.136, the Fractional-Logit formulation requires 21.97 seconds on average, compared with 4.95 seconds for the bounded nonlinear regression formulation. It therefore requires 4.44 times the mean runtime of the bounded nonlinear regression formulation. This setting is particularly relevant because it belongs to the first tested approximation region in which the Fractional-Logit approximation reaches or exceeds the predictive accuracy of the bounded nonlinear regression function.

The A_6_3 instances provide a more demanding comparison. The bounded nonlinear regression formulation obtains OPTIMAL status for all ten instances, requiring 970.71 seconds on average. Across the 70 Fractional-Logit instance–approximation configurations, 52 obtain OPTIMAL status and 18 reach the 7,200-second time limit. In every one of these 18 time-limit configurations, the corresponding bounded nonlinear regression formulation obtains OPTIMAL status.

Moreover, in each of the 52 configurations for which both formulations obtain OPTIMAL status, the Fractional-Logit formulation is slower than the corresponding bounded nonlinear regression formulation. Pooling these 52 jointly optimal configurations, the bounded nonlinear regression and Fractional-Logit formulations require 644.86 and 3,642.15 seconds on average, respectively. Thus, the Fractional-Logit formulation requires 5.65 times the mean runtime of the bounded nonlinear regression formulation.

At the threshold-near error of 0.136, both formulations obtain OPTIMAL status for eight instances. On this jointly optimal subset, the bounded nonlinear regression and Fractional-Logit formulations require 558.82 and 4,165.81 seconds on average, respectively. The Fractional-Logit formulation therefore requires 7.45 times the mean runtime of the bounded nonlinear regression formulation. For the other two instances, the bounded nonlinear regression formulation obtains OPTIMAL status, whereas the Fractional-Logit formulation reaches the 7,200-second limit.

At the finer error of 0.010, both formulations obtain OPTIMAL status for nine instances. On these instances, their respective mean runtimes are 853.75 and 2,187.82 seconds, corresponding to a runtime ratio of 2.56. For the remaining instance, only the bounded nonlinear regression formulation obtains

OPTIMAL status. Across all 18 time-limited Fractional-Logit configurations, the remaining MIP gap averages 4.03% and ranges from 1.08% to 9.13%.

The computational time is not monotonic in the number of approximation segments, reflecting the combined influence of formulation structure, relaxation, and branch-and-bound search. Nevertheless, the results show a consistent pattern. The modest predictive improvement provided by Fractional Logit comes at a substantially higher computational cost after the function is embedded into O-CAM. When both formulations obtain OPTIMAL status, the Fractional-Logit formulation typically requires several times the computation of the bounded nonlinear regression formulation; for a further subset of instances, the bounded nonlinear regression formulation obtains OPTIMAL status while the Fractional-Logit formulation cannot do so within 7,200 seconds.

0.4 Compatibility and Computational Effects within PAD^2

We further embed two representative Fractional-Logit approximations, with requested errors of 0.136 and 0.010, into both the linear-predictor and nonlinear-predictor variants of PAD^2. The experimental design contains two instance groups, ten instances per group, two PAD^2 predictor variants, and three acceptance-function specifications, producing 120 completed PAD^2 tasks.

For each instance, the runtime of the Fractional-Logit implementation is divided by that of the corresponding bounded nonlinear regression implementation under the same instance group and PAD^2 predictor variant. The resulting ratios are then averaged across the ten paired instances.

For A_4_2, the mean paired time ratios for the 0.136 and 0.010 Fractional-Logit approximations are 3.22 and 3.29, respectively, for the linear-predictor variant, and 1.54 and 1.51 for the nonlinear-predictor variant. For A_6_3, the corresponding ratios are 1.96 and 1.92 for the linear-predictor variant and 1.36 and 1.20 for the nonlinear-predictor variant.

All 120 PAD^2 tasks are completed successfully, confirming that the Fractional-Logit specification can be incorporated into PAD^2 without changing the overall algorithmic workflow. However, the alternative specification consistently introduces additional computational cost. For the linear-predictor variant, the mean paired time ratios range from 1.92 to 3.29, whereas for the nonlinear-predictor variant they range from 1.20 to 1.54. Overall, embedding Fractional Logit increases PAD^2 computation time by approximately 20%–229%, depending on the instance group and predictor variant.

Taken together, the results show that exact Fractional Logit provides only a modest improvement in predictive fit relative to the bounded nonlinear regression specification, reducing validation MSE by 5.35% and increasing R^2 by approximately 0.015. Retaining comparable Fractional-Logit accuracy inside the optimization framework, however, introduces substantial additional computational cost in O-CAM and increases PAD^2 computation time by approximately 20%–229%. These findings support retaining the bounded nonlinear regression specification as an accuracy–tractability choice for the real-time optimization framework.